

MAT 549 HW07

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Problem 1 (17-1). Let M be a smooth manifold with or without boundary and let $\omega \in \Omega^p(M)$, $\eta \in \Omega^q(M)$ be closed forms. Show that the de Rham cohomology class of $\omega \wedge \eta$ depends only on the cohomology classes of ω and η , and thus there is a well-defined bilinear map $\smile: H_{dR}^p(M) \times H_{dR}^q(M) \rightarrow H_{dR}^{p+q}(M)$, called the cup product, given by $[\omega] \smile [\eta] = [\omega \wedge \eta]$.

Solution. Let $\omega' \in [\omega]$ and $\eta' \in [\eta]$, so $\omega' - \omega = d\alpha$ and $\eta' - \eta = d\beta$ are exact. Then

$$\omega \wedge \eta - \omega' \wedge \eta' = (\omega - \omega') \wedge \eta + \omega' \wedge (\eta - \eta') = d\alpha \wedge \eta + \omega' \wedge d\beta = d(\alpha \wedge \eta + \omega' \wedge \beta),$$

where the last equality uses the fact that η and ω' are closed, so $[\omega \wedge \eta] = [\omega' \wedge \eta']$. Thus $\smile$ is a well-defined map on the cohomology groups, and it inherits bilinearity from the bilinearity of $\wedge$: explicitly, we can write

$$[a\omega + b\eta] \smile [\alpha] = [(a\omega + b\eta) \wedge \alpha] = [a\omega \wedge \alpha] + [b\eta \wedge \alpha] = a[\omega] \smile [\alpha] + b[\eta] \smile [\alpha]$$

to show linearity in the first argument, and essentially the same calculation shows linearity in the second argument. $\square$

Problem 2 (17-12). Suppose M and N are compact, connected, and oriented n -manifolds, and $F: M \rightarrow N$ is a smooth map. Prove that if $\int_M F^* \eta \neq 0$ for some $\eta \in \Omega^n(N)$, then F is surjective. Give an example to show that F can be surjective even if $\int_M F^* \eta = 0$ for every $\eta \in \Omega^n(N)$.

Solution. Suppose F is not surjective. Then, any point not in $F(M)$ is vacuously a regular value for F , so the degree of F is zero; that is,

$$\int_M F^* \eta = \deg F \cdot \int_N \eta = 0$$

for all $\eta \in \Omega^n(N)$.

To show that the converse does not hold, let $M = N = S^1 \cong \mathbb{R}/\mathbb{Z}$ and consider $F: S^1 \rightarrow S^1$ given by

$$F(x) = \sin^2(\pi x).$$

Since $x \mapsto \sin^2(\pi x)$ is a surjective smooth map $\mathbb{R} \rightarrow [0, 1]$, it descends to the smooth map $F: S^1 \rightarrow S^1$. To see that F has degree 0 we note that it is homotopic to a constant map via the homotopy

$$\begin{aligned} H: S^1 \times [0, 1] &\rightarrow S^1 \\ (x, t) &\mapsto tF(x) \end{aligned}$$

(thinking of the S^1 in the codomain as $[0, 1]$ with its endpoints identified). $\square$

Problem 3 (17-13). Let $T^2 = S^1 \times S^1$ be the 2-torus. Consider the two maps $f, g: T^2 \rightarrow T^2$ given by $f(w, z) = (w, z)$ and $g(w, z) = (z, \bar{w})$. Show that f and g have the same degree, but are not homotopic.

Solution. Since f is the identity map, it has degree 1. To show that g has degree 1 observe that g is a diffeomorphism because it has a smooth inverse $g^{-1}(w, z) = (\bar{z}, w)$, and it is orientation-preserving because the determinant of the differential of g at any point is

$$\begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} = 1,$$

and we know that orientation-preserving diffeomorphisms have degree 1. However, these maps are not homotopic because f induces the identity homomorphism on $\pi_1(T^2)$ while g induces the homomorphism

$$g_*([\gamma_1], [\gamma_2]) = ([\gamma_2], -[\gamma_1])$$

which is clearly not the identity, where $(\gamma_1, \gamma_2): S^1 \rightarrow T^2$ is an arbitrary loop in T^2 . □

Problem 4. Find the de Rham cohomology groups of the 3-torus $T^3 = S^1 \times S^1 \times S^1$.

Solution. We more generally compute $H^k(T^n) \cong \mathbb{R}^{\binom{n}{k}}$ for all nonnegative integers n and integers k by induction on n , starting with the observation that the singleton set T^0 has de Rham cohomology

$$H^k(T^0) = \begin{cases} \mathbb{R} = \mathbb{R}^{\binom{0}{0}} & \text{if } k = 0 \\ 0 & \text{if } k > 0. \end{cases}$$

Write $S^1 = U \cup V$ where U and V are arcs of S^1 and $U \cap V = S^1$. Then, the intersection $U \cap V$ consists of two disjoint arcs. The sets $\tilde{U} = U \times T^{n-1}$ and $\tilde{V} = V \times T^{n-1}$ give us an open cover of T^n , and the corresponding Mayer-Vietoris sequence tells us that

$$\dots \rightarrow H^{k-1}(\tilde{U} \cap \tilde{V}) \xrightarrow{\delta^{k-1}} H^k(T^n) \xrightarrow{\alpha^k} H^k(\tilde{U}) \oplus H^k(\tilde{V}) \xrightarrow{\beta^k} H^k(\tilde{U} \cap \tilde{V}) \rightarrow \dots$$

is exact for all k . Since U and V are contractible, $\tilde{U}$ and $\tilde{V}$ are homotopy equivalent to T^{n-1} . Also, $\tilde{U} \cap \tilde{V}$ has two components W_1, W_2 that are homotopy equivalent to T^{n-1} . Writing $H^k(\tilde{U} \cap \tilde{V}) \cong H^k(W_1) \oplus H^k(W_2)$, the map β^k is given by

$$\eta^k([\omega], [\eta]) = ([\omega|_{W_1}] - [\eta|_{W_1}], [\omega|_{W_2}] - [\eta|_{W_2}]),$$

so the image of this map, which equals the kernel of δ^k , is the diagonal $([\omega], [\omega])$. Observe also that the image of α^k (which is also the kernel of β^k) is also the diagonal of $H^k(T^{n-1}) \oplus H^k(T^{n-1})$, which is isomorphic to the image of β^k . Thus the image and kernel of β^k have the same dimension $\binom{n-1}{k}$, using the inductive hypothesis.

Finally, the dimension of $H^k(T^n)$ is sum of the dimension of the image of α^k and the dimension of the image of δ^{k-1} ; the former we know to be $\binom{n-1}{k}$, and the latter is $\dim H^{k-1}(\tilde{U} \cap \tilde{V}) = 2\binom{n-1}{k-1}$ minus $\dim \ker \delta^{k-1} = \dim \operatorname{im} \beta^{k-1} = \binom{n-1}{k-1}$, meaning

$$\dim H^k(T^n) = \binom{n-1}{k} + \binom{n-1}{k-1} = \boxed{\binom{n}{k}}.$$

In the $n = 3$ case this means

$$H^k(T^3) = \begin{cases} \mathbb{R} & \text{if } k = 0, 3 \\ \mathbb{R}^3 & \text{if } k = 1, 2 \\ 0 & \text{if } k > 3. \end{cases} \quad \square$$